Temă – lab. 13

30 el. $\begin{cases} \text{group 1: } [3, 5, 3, 2, 4, 6, 9, 3, 8, 10] \to 10 \text{ el.} \\ \text{group 2: } [10, 8, 15, 9, 11, 16, 17, 15, 7, 10] \to 10 \text{ el.} \\ \text{group 3: } [29, 15, 14, 15, 17, 10, 8, 11, 18, 19] \to 10 \text{ el.} \end{cases}$

Ex. 1: Apply One-Way ANOVA and post-hoc analysis (the Turkey's method or the Scheffe approach) on data.

One-Way Analysis:

$H_0 : \overbrace{\mu_1 = \mu_2 = \mu_3}^{\text{the means are equal}}$

$$sum_{gr.1} = 3+5+3+2+4+6+9+3+8+10$$
$$= 53$$

$$sum_{gr.2} = 10+8+15+9+11+16+17+15+7+10$$
$$= 118$$

$$sum_{gr.3} = 29+15+14+15+17+10+8+11+18+19$$
$$= 156$$

The grand mean : $\overline{x}$

$$\overline{x} = \frac{\sum x_i}{N} = \frac{sum\ gr_1 + sum\ gr_2 + sum\ gr_3}{10+10+10}$$

$$= \frac{53+118+156}{30} = \frac{327}{30} = 10{,}9$$

The Sum of Squares for the treatment term : SS between

$$SS_{between} = S_1^2 + S_2^2 + S_3^2 \text{ OR}$$
$$= \sum n_j (\overline{x_j} - \overline{x})^2$$

$$\text{sum}_{gr.1} = 53 \Rightarrow \overline{X}_1 = \frac{53}{10} = 5{,}3$$

$$\text{sum}_{gr.2} = 118 \Rightarrow \overline{X}_2 = \frac{118}{10} = 11{,}8$$

$$\text{sum}_{gr.3} = 156 \Rightarrow \overline{X}_3 = \frac{156}{10} = 15{,}6$$

$$\begin{aligned} SS_{between} &= 10(5{,}3 - 10{,}9)^2 + \\ &\quad 10(11{,}8 - 10{,}9)^2 + \\ &\quad 10(15{,}6 - 10{,}9)^2 \\ &= 10 \cdot [(-5{,}6)^2 + (0{,}9)^2 + 4{,}7^2] \\ &= 10 \cdot (31{,}36 + 0{,}81 + 22{,}09) \\ &= \underline{\underline{542{,}6}} \end{aligned}$$

SS_{error} or the SS_{within}:

$$\begin{aligned} SS_{within}(gr.1) &= 3(3 - 5{,}3)^2 + (5 - 5{,}3)^2 + (2 - 5{,}3)^2 \\ &\quad + (4 - 5{,}3)^2 + (6 - 5{,}3)^2 + (9 - 5{,}3)^2 \\ &\quad + (8 - 5{,}3)^2 + (10 - 5{,}3)^2 \\ &= 3 \cdot (-2{,}3)^2 + (-0{,}3)^2 + (-3{,}3)^2 + (-1{,}3)^2 + \\ &\quad + (0{,}7)^2 + (3{,}7)^2 + (2{,}7)^2 + (4{,}7)^2 \\ &= 15{,}87 + 0{,}09 + 10{,}89 + 1{,}69 + 0{,}49 + \\ &\quad + 13{,}69 + 7{,}29 + 22{,}09 \\ &= \underline{\underline{72{,}1}} \end{aligned}$$

$$SS_{within}\ (\text{gr. } 2) = 2(10-11{,}8)^2 + (8-11{,}8)^2 + 2(15-11{,}8)^2 + (9-11{,}8)^2 + (11-11{,}8)^2 + (16-11{,}8)^2 + (17-11{,}8)^2 + (7-11{,}8)^2$$

$$= 2(-1{,}8)^2 + (-3{,}8)^2 + 2(3{,}2)^2 + (-2{,}8)^2 + (-0{,}8)^2 + (4{,}2)^2 + (5{,}2)^2 + (-4{,}8)^2$$

$$= 6{,}48 + 14{,}44 + 20{,}48 + 7{,}84 + 0{,}64 + 17{,}64 + 27{,}04 + 23{,}04 = \underline{117{,}56}$$

$$SS_{within}\ (\text{gr. } 3) = 2(15-15{,}6)^2 + (14-15{,}6)^2 + (17-15{,}6)^2 + (10-15{,}6)^2 + (8-15{,}6)^2 + (11-15{,}6)^2 + (18-15{,}6)^2 + (19-15{,}6)^2 + (29-15{,}6)^2$$

$$= 2(-0{,}6)^2 + (-1{,}6)^2 + (1{,}4)^2 + (-5{,}6)^2 + (-7{,}6)^2 + (-4{,}6)^2 + (2{,}4)^2 + (3{,}4)^2 + (13{,}4)^2$$

$$= 0{,}72 + 2{,}56 + 1{,}96 + 31{,}36 + 57{,}76 + 21{,}16 + 5{,}76 + 11{,}56 + 179{,}56$$

$$= \underline{312{,}4}$$

$$SS_{within} = 72{,}1 + 117{,}56 + 312{,}4 = \underline{502{,}06}$$

The mean square between: MS_b

$$MS_b = \frac{SS_{between}}{k-1}, \quad k-1 = \text{the degrees of freedom } (3 \text{ gr.} - 1)$$

$$= \frac{542{,}6}{3-1} = \frac{542{,}6}{2} = \underline{271{,}3}$$

The mean square within (error): MS_w

$$MS_w = \frac{SS_{within}}{N-K}, \quad df_{error} = df_w$$

$= df_{error} = 30 - 3 = 27$
(30 people − 3 gr.)

$$= \frac{502,06}{27}$$

$$= \underline{\underline{18,59}}$$

F − statistic: F observed

$$F_{observed} = \frac{MS_b}{MS_w} = \frac{271,3}{18,59} = \underline{\underline{14,59}}$$

	SS	df	MS	F obs
Beetween	542,6	2	271,3	14,59
Within	502,06	27	18,59	
Total	1044,66	29		

F critical: $\alpha = 0,05$; $df = 2$; $df_w = 27$

$\Rightarrow F_{critical} = \underline{\underline{3,354}}$

$F_{observed} > F_{critical}$ $(14,59 > 3,354)$

$\Rightarrow$ reject the null hypothesis
$(\mu_0 = \mu_1 = \mu_2)$

Tukey and Scheffe post-hoc tests

threshhold

The formula for the Tukey's HSD post-hoc test:

$$\frac{\bar{x}_1 - \bar{x}_2}{\sqrt{MS_W \left(\frac{1}{n_1}\right)}}$$

$n_1 = n_2 = n_3 = 10$

$\bar{x}_1 = 5,3$; $\bar{x}_2 = 11,8$; $\bar{x}_3 = 15,6$

$MS_W = 18,59$

Critical value for $k = 3$: $3,51$

Gr. 1 vs. Gr. 2	Gr. 1 vs Gr. 3	Gr. 2 vs Gr. 3
$\frac{\lvert 5.3 - 11,8 \rvert}{\sqrt{\frac{18,59}{10}}}$	$\frac{\lvert 5,3 - 15,6 \rvert}{\sqrt{\frac{18,59}{10}}}$	$\frac{\lvert 11,8 - 15,6 \rvert}{\sqrt{\frac{18,59}{10}}}$
$\frac{6,5}{\sqrt{1,859}}$	$\frac{10,3}{\sqrt{1,859}}$	$\frac{3,8}{\sqrt{1,859}}$
$\frac{6,5}{1,363}$	$\frac{10,3}{1,363}$	$\frac{3,8}{1,363}$
$4,768 > 3,51$	$7,556 > 3,51$	$2,787 \leq 3,51$
$\Rightarrow$ reject the H_1 $(\bar{x}_1 = \bar{x}_2)$	$\Rightarrow$ rejects the H_2 $(\bar{x}_1 = \bar{x}_3)$	$\Rightarrow$ fails to reject H_3 $(\bar{x}_2 = \bar{x}_3)$

Minimum significant difference for Scheffe test:

$$\sqrt{(k-1)\cdot F_{critical}\cdot MS_w\cdot\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}$$

$$=\sqrt{(3-1)\cdot 3{,}35\cdot 18{,}59\cdot \frac{2}{10}} = \sqrt{24{,}9106} = 4{,}991$$

Gr.1 vs. Gr.2	Gr.1 vs Gr.3	Gr.2 vs Gr.3
$\lvert 5.3 - 11{,}8\rvert$	$\lvert 5{,}3-15{,}6\rvert$	$\lvert 11{,}8-15{,}6\rvert$
$6{,}5 > 4{,}991$ $\Rightarrow$ Reject the H_1 $(\overline{x}_1 = \overline{x}_2)$	$10{,}3 > 4{,}991$ $\Rightarrow$ Reject the H_2 $(\overline{x}_1 = \overline{x}_3)$	$3{,}8 \leq 4{,}991$ $\Rightarrow$ fail to reject the H_3 $(\overline{x}_2 = \overline{x}_3)$

Scheffe Grouping	Mean	N	Group
A	15,6	10	3
A	11,8	10	2
B	5,3	10	1

For both Tukey and Scheffe

- rejected H_1 $(\overline{x}_1 = \overline{x}_2)$
- rejected H_2 $(\overline{x}_1 = \overline{x}_3)$
- fail to reject H_3 $(\overline{x}_2 = \overline{x}_3)$

Confidence Intervals:

t-value = $n-1 = 9$ (two-tailed test for $\alpha = 0{,}05$)
$\rightsquigarrow 2{,}262$

Standard error = $SE = \sqrt{\frac{MSw}{n}} = \sqrt{\frac{18{,}59}{10}} = 1{,}3637$

Margin of error = $ME = t \cdot SE$

$= 2{,}262 \cdot 1{,}3637$

$= 3{,}0846$

Ranges: gr_1: $5{,}3 \pm 3{,}0846$

$\Rightarrow 2{,}2154 \longleftrightarrow 8{,}3846$

gr_2: $11{,}8 \pm 3{,}0846$

$\Rightarrow 8{,}7154 \longleftrightarrow 14{,}8846$

gr_3: $15{,}6 \pm 3{,}0846$

$\Rightarrow 12{,}5154 \longleftrightarrow 18{,}6846$

Group	is it in interval?
1	$\overline{X_2}$: 11,8 is outside the range
	$\overline{X_3}$: 15,6 is outside the range
2	$\overline{X_1}$: 5,3 is outside the range
	$\overline{X_3}$: 15,6 is outside the range
3	$\overline{X_1}$: 5,3 is outside the range
	$\overline{X_2}$: 11,8 is outside the range